Hierarchical Deep Learning for Zenith-to-Horizon Electron Trajectory Reconstruction in the DAMPE BGO Calorimeter Hierarchical Deep Learning for BGO-Only Electron Track Reconstruction with DAMPE

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Abstract

Reliable track reconstruction is fundamental to the Dark Matter Particle Explorer (DAMPE) mission, as the determination of multiple cosmic ray properties relies heavily on precise trajectory information. Conventional reconstruction algorithms focus primarily on penetrating events, effectively leveraging measurements from the Silicon-Tungsten Tracker (STK) and the BGO calorimeter, with the STK providing the critical anchor for track precision. However, for large-angle incident events—particularly those traversing the BGO calorimeter quasi-horizontally—the absence of reliable STK measurements renders traditional methods ineffective. To unlock DAMPE's observational potential at large angles, specifically for electron-like cosmic rays, we propose a hierarchical deep learning framework relying solely on BGO shower profiles in this work. To ensure robust application to unlabeled flight data, we introduce a Support Vector Machine (SVM) classifier to reliably distinguish between standard-angle (0°–60°) and large-angle (60°–90°) regimes based on distinct shower morphologies. Accordingly, we develop specialized models adapted to each interval: a physics-guided hybrid model fusing the DAMPE energy-weighted Center-of-Gravity reconstruction with deep learning for standard angles, and a deep-learning-dominant strategy to

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recover trajectories from truncated showers in the STK-blind region. Validated on Monte Carlo datasets, our method achieves nearly 100% reconstruction efficiency across the full angular range, with an angular resolution (68% containment radius) of 1.40° for standard angles and 1.31° for large angles. A cross-check on flight electron candidates with STK tracks (30°–71°) shows degree-level agreement between the BGO-only predictions and the STK reference directions (median angular difference 1.44°, 68th percentile 1.98°). By extending reliable reconstruction into the 60°–90° band where the STK is unavailable, this work raises the effective acceptance of the detector by a factor of 1.8.

Keywords: Particle Track Reconstruction, Deep Learning, BGO Calorimeter, Large-Angle Events

1. Introduction

The Dark Matter Particle Explorer (DAMPE), launched in December 2015, is a high-energy cosmic-ray and gamma-ray observatory operating in a sun-synchronous orbit [1]. Its primary scientific objectives include the indirect search for dark matter, the precise measurement of cosmic-ray spectra (electrons [2], protons [3], helium [4, 5], and heavier nuclei [6]) up to the PeV scale, and high-energy gamma-ray astronomy [7]. To fulfill the stringent requirements for precise energy measurement and particle identification, the DAMPE payload comprises four sub-detectors: a Plastic Scintillator Detector (PSD) [8, 9, 10, 11], a Silicon-Tungsten Tracker (STK) [12, 13, 14], a Bismuth Germanium Oxide (BGO) imaging calorimeter [15, 16, 17], and a Neutron Detector (NUD) [18].

In the standard reconstruction pipeline adopted by the DAMPE collaboration, precise trajectory reconstruction relies heavily on the synergy between the BGO calorimeter and the STK [19], a design philosophy shared by major space-borne experiments such as Fermi-LAT [20] and CALET [21, 22]. Typically, the shower axis is first coarsely estimated using the BGO energy distribution, which then seeds the search for track candidates in the STK. The STK, with its fine silicon strips, supplies the precise angular measurement that, together with the BGO seed, yields a trajectory resolution of approximately 0.2° at 10 GeV for the combined BGO+STK pipeline [12]. This trajectory is then projected back to the PSD for absolute charge measurement. Although this multi-detector synergy yields high precision, it suffers

from an intrinsic geometric limitation: the incident particle must traverse the fiducial volumes of the PSD, STK, and BGO simultaneously.

Due to the limitations of the aspect ratio and trigger logic of the DAMPE payload [23], events with zenith angles exceeding $\sim 60^\circ$ often bypass the fiducial volumes of the PSD and STK, or traverse only limited layers of STK, which is insufficient to form a track candidate. Consequently, the standard Kalman-filter-based tracking algorithms become inapplicable [19]. For scientific goals that require high statistics, such as the search for monochromatic gamma-ray lines [24] or the precise measurement of rare high-energy spectral features, maximizing the effective field of view is paramount. Therefore, the development of a trajectory reconstruction method that relies exclusively on the BGO calorimeter, the key sub-detector with a sufficiently large acceptance to capture these events, is essential.

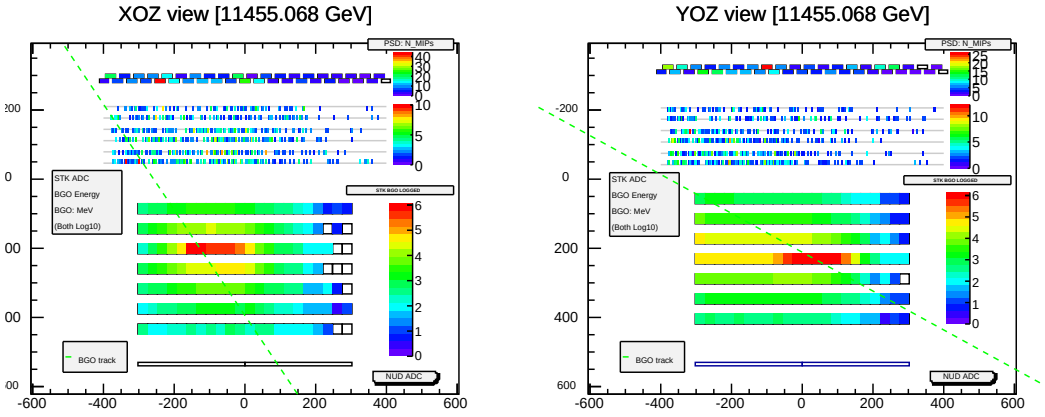


Figure 1: Schematic of the geometric limitation. The green dashed line illustrates the BGO-only track reconstructed by the Standard BGO Algorithm, representing a large-angle trajectory that traverses the BGO calorimeter but bypasses the PSD and STK detectors.

Reconstructing trajectories using only the BGO calorimeter is difficult, particularly for large-angle events. The standard BGO reconstruction is the official DAMPE energy-weighted Center-of-Gravity (CoG) track reconstruction [25, 26], hereafter the Standard BGO Algorithm; it reconstructs the shower axis from the energy-weighted centroids of the BGO layers. While effective for showers that are fully contained within the calorimeter, this analytical approach degrades once the incident angle grows large, as shown in Figure 1. As the incident angle increases, the electromagnetic or hadronic

showers are often not fully contained within the calorimeter depth. This phenomenon, known as "side leakage," causes a systematic shift in the energy-weighted centroid of the shower development, leading to substantial biases in the reconstructed direction. In addition, the limited number of BGO layers traversed by large-angle particles reduces the available information for fitting, rendering the CoG track fit unstable and unreliable.

Deep learning has been widely applied with success to calorimeter shower analysis in high-energy physics [27, 28, 29, 30, 31, 32, 33, 34]. For the DAMPE experiment, Tykhonov et al. [35] successfully applied Convolutional Neural Networks (CNNs) to improve the BGO shower axis prediction, providing a superior seed for the STK. However, existing works have largely focused on optimizing performance within the standard geometric acceptance, that is, improving the resolution of events that can be observed by the STK, rather than extending reconstruction into the large-angle regime where the STK is unavailable. The present method instead operates on BGO data alone to recover events in the 60°–90° band that the STK-dominated pipeline discards, complementing rather than competing with STK-based reconstruction.

The approach in this work tailors the reconstruction logic to the physical shower morphology at each incident angle. For the standard-angle regime, we embed physical principles and geometric features directly into the neural network architecture, ensuring that the model adheres to known physical laws during the learning process. This physics-informed feature engineering coupled with a physics-gated hybrid fusion can be viewed as a "gray-box" model [36, 37, 38, 39]: it keeps predictions consistent with known physical constraints while retaining the representational power of deep learning. For the large-angle regime, where physical priors become unreliable due to severe side leakage, we adopt a Deep-Learning-Dominant(DLD) strategy that relies on enhanced input representation and multi-task learning to recover trajectory information from truncated showers. We further developed a Support Vector Machine (SVM) [40] classifier to automatically categorize events into "standard-angle" (0°–60°) and "large-angle" (60°–90°) regimes based on shower morphology, thereby enabling the processing of unlabeled flight data.

The organization of the paper is as follows: Section 2 presents the simulation data, Section 3 introduces and implements the methods, Section 4 covers performance evaluation, and discussion and conclusion are provided in Sections 5 and 6, respectively.

2. Data and Simulation

The DAMPE detector consists of four sub-detectors. Among these, the BGO calorimeter is the primary instrument for energy measurement and is central to electron-hadron discrimination. It is composed of 308 BGO crystal bars, each measuring $2.5 \times 2.5 \times 60$ cm^3. These bars are arranged in 14 layers, with each layer containing 22 bars [1, 16]. The layers are stacked in an orthogonal hodoscopic configuration, in which odd-numbered layers are oriented along the Y-axis and even-numbered layers along the X-axis. This design provides a total depth of approximately 31 radiation lengths and allows for three-dimensional tomographic reconstruction of the particle shower development. The geometric arrangement of the BGO bars naturally maps to a 14×22 grid, which is the foundational data structure for the deep learning model.

To train and validate the proposed deep learning framework, a large-scale dataset was generated using the GEANT4 [41] simulation toolkit. The simulation incorporates a detailed model of the DAMPE detector geometry [19] and material properties, along with extensive physics lists for particle interactions [42, 43, 44]. The dataset focuses on electron events, which are the primary targets of this work and can be separated from hadronic showers using BGO data alone. The detailed simulation parameters are configured as follows:

- **Particle Type**: Electrons (e^-).
- **Energy Range**: 100 GeV to 1 TeV, following a power-law spectrum with a spectral index of -1.0 to ensure sufficient statistics at high energies.
- **Angular Distribution**: The incident particles were generated on a sphere of radius 1.38 m centred on the detector axis, 24.7 cm above the geometric centre of the BGO calorimeter, with an isotropic angular distribution covering the full zenith angle range from 0° to 90°; this radius is referred to as the generation radius. This full-angle coverage is essential for training the model to handle large-angle events that are typically discarded by standard selection criteria.
- **Total Statistics**: A total of 11,894,154 events were selected. The dataset was randomly split into training (80%) and testing (20%) sets.

The two expert branches are trained on overlapping angular sub-ranges of this sample (0°–70° and 50°–90°); the 80/20 train/test partition is unchanged.

To facilitate efficient learning by the neural network, we perform a series of pre-processing steps to construct two distinct input streams: the BGO Energy Map and the Physics-informed Feature Vector. The energy deposition in the 308 BGO bars is mapped into a 14×22 matrix, in which each element (i, j) represents the energy deposited in the j-th bar of the i-th layer. Although this projection flattens the three-dimensional detector geometry into a two-dimensional grid, the orthogonal hodoscopic nature of the calorimeter is implicitly preserved in the row-wise structure (alternating X and Y orientations). This allows the model to learn spatial correlations specific to the cross-layered topology of the detector. To handle the large dynamic range of energy deposits (spanning from GeV to TeV) and to normalize the input distribution, we construct a dual-channel input tensor of shape $(14, 22, 2)$:

1. **Normalized Energy Channel**: To capture the relative shower shape independent of the absolute energy scale, the energy in each bar is normalized by the total energy deposited in the event:

$$E_{\text{norm}}^{(i,j)} = \frac{E^{(i,j)}}{\sum_{i,j} E^{(i,j)}} \tag{1}$$

2. **Log-transformed Energy Channel**: To preserve information about low-energy deposits tracing the shower periphery while compressing the high-energy core, we apply a logarithmic transformation:

$$E_{\log}^{(i,j)} = \ln(1 + E^{(i,j)}) \tag{2}$$

In addition to the raw energy map, we extract a 24-dimensional vector of physics-based features to guide the network. These features encapsulate domain knowledge about electromagnetic shower development and provide explicit physical guidance to the network. The features are categorized as follows:

1. **Layer-wise Energy Distribution (7 features)**: The total energy deposited in each of the 7 X-Y layer pairs. This describes the longitudinal profile of the shower.
2. **Layer-wise Maximum Energy (7 features)**: The average of the maximum energy deposition in the X and Y bars for each layer pair, indicating the intensity of the shower core.

3. **Layer-wise Maximum-Energy Bar Position (7 features)**: For each layer pair, the average bar index of the X and Y bars that record the maximum energy deposition, providing a coarse estimate of the shower axis in that layer.
4. **Global Energy-Weighted Centroid (3 features)**: The energy-weighted mean position (X_c, Y_c, Z_c) of the entire shower in the detector coordinate system. This is calculated as:

$$X_c = \frac{\sum_{i,j} x_{i,j} \cdot E_{i,j}}{\sum_{i,j} E_{i,j}}, \quad Y_c = \frac{\sum_{i,j} y_{i,j} \cdot E_{i,j}}{\sum_{i,j} E_{i,j}}, \quad Z_c = \frac{\sum_{i,j} z_{i,j} \cdot E_{i,j}}{\sum_{i,j} E_{i,j}} \tag{3}$$

   where $(x_{i,j}, y_{i,j}, z_{i,j})$ are the spatial coordinates of the j-th bar in the i-th layer.

These 24 features serve as explicit physical constraints for the standard-angle branch; the exact composition of the physics vector is regime-specific, and the large-angle variant is detailed in Section 3.4.

3. Methodology

We now detail the architecture of this hierarchical framework.

3.1. Overall Framework

Electromagnetic showers in the BGO calorimeter develop differently depending on the incident angle. To effectively address these variations, we categorize incident events into two distinct regimes: the standard-angle regime (0°–60°) and the large-angle regime (60°–90°), also referred to as standard and large incidence. This categorization is empirically determined by two independent facts that happen to coincide near 60°: the geometric acceptance of the DAMPE detector, which removes the STK from the measurement above this angle, and the degradation of the Standard BGO Algorithm at large incidence. Monte Carlo simulations demonstrate that the reconstruction efficiency of the Standard BGO Algorithm remains acceptable for zenith angles up to approximately 60°, but drops steeply beyond this threshold (Section 4.2). The two are physically distinct. Above 60° the shower is frequently not contained within the calorimeter depth, so side leakage biases the energy-weighted centroid and the CoG fit degrades; independently, particles at these angles typically bypass the fiducial volume of the STK, leaving

the BGO calorimeter as the sole source of trajectory information. Consequently, events in the standard-angle regime are typically well-contained, exhibiting clear longitudinal and lateral profiles. Conversely, events in the large-angle regime suffer from severe side leakage and truncated path lengths, producing truncated, irregular shower topologies that require a specialized deep-learning strategy.

To accommodate these distinct physical regimes, we design a two-branch pipeline:

1. **Zenith Categorization**: An incoming event is first processed by SVM classifier to obtain its approximate incident-angle range.
2. **Model Selection**: Based on the above classification, the event is routed to either the standard-angle Branch (utilizing a physics-guided hybrid strategy) or the large-angle Branch (utilizing a deep-learning-dominant strategy), irrespective of the wider angular range used in training.

This hierarchical structure ensures that the reconstruction algorithm is optimal for the specific topology of the event, balancing robustness and precision.

3.2. Angle Classifier

Since flight data lack ground-truth labels for incident angles, an automatic categorization mechanism is a prerequisite for the hierarchical framework. We developed a classifier based on a Support Vector Machine (SVM) [40, 45] to distinguish between standard-angle and large-angle events from shower morphology alone.

Input representation. Rather than feeding the full 308-dimensional raw energy vector, the classifier operates on a compact 7-dimensional physics feature vector extracted from the 14×22 BGO energy matrix. Writing $E_{l,b}$ for the energy in bar b of layer l and $E_l = \sum_b E_{l,b}$ for the layer energy, the vector combines three geometric descriptors with four basic shower quantities:

- **Longitudinal (depth) RMS** (σ_{long}, in *layer* units) — the energy-weighted Root Mean Square (RMS) spread of the layer index:

$$\sigma_{\text{long}} = \Big[\sum_l (l-\bar{l})^2 E_l \Big/ \sum_l E_l\Big]^{1/2}, \quad \bar{l} = \sum_l l E_l \Big/ \sum_l E_l \tag{4}$$

- **Lateral RMS width** (σ_{lat}, in *bar* units) — the average over the 14 layers of the energy-weighted RMS spread of the bar index within each layer:

$$\sigma_{\mathrm{lat}} = \tfrac{1}{14} \sum_l \left[\sum_b (b - \bar{b}_l)^2 E_{l,b} \Big/ \sum_b E_{l,b} \right]^{1/2}, \quad \bar{b}_l = \sum_b b E_{l,b} \Big/ \sum_b E_{l,b} \tag{5}$$

- **Shower-axis slope magnitude** (s_{mag}, in *bar units per layer*, numerically close to the tangent of the shower tilt) — the energy-weighted linear regression of each layer's transverse centroid against the layer index, performed separately for the X-view (even layers) and the Y-view (odd layers) and combined as

$$s_{\mathrm{mag}} = \sqrt{s_x^2 + s_y^2} \tag{6}$$

- **Four basic quantities** — the layer of maximum energy deposition, the total deposited energy, and the two individual slope projections $|s_x|$ and $|s_y|$.

All quantities are expressed in the detector's native bar/layer indices rather than physical distances, keeping them scale-free and directly observable in unlabeled flight data, and together they capture the essential morphological differences between standard and large-angle showers.

Model and training. We use a scikit-learn Support Vector Classifier(SVC) with an Radial Basis Function(RBF) kernel ($C = 10$, $\gamma =$ `scale`), preceded by a `StandardScaler` for z-score normalization of the input features. The classifier is trained on a class-balanced sample of 50,000 Monte Carlo events drawn from a stratified pool of 300,000 events, using an 80/20 train/test partition; events are labeled by the true incident-angle threshold $\theta > 60°$. On the held-out test set the classifier reaches an accuracy of 98.4%.

Characterization and failure modes. Table 1 reports the per-class **precision**, **recall**, and **F1 score**; both regimes are well balanced. The two metrics answer complementary questions and differ only in their denominator. Taking the large-angle class as an example, let TP be the number of genuinely large-angle events that the classifier also labels large-angle, FP the number of standard-angle events it mislabels as large-angle, and FN the number of genuinely large-angle events it mislabels as standard-angle. **Precision** = TP/(TP + FP) asks, *of all events the classifier flags as large-angle, how many really are?* It measures the *purity* of the routing. **Recall**

$= \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$ asks, *of all genuinely large-angle events, how many does the classifier find?* It measures the *completeness*. The **F1 score** is their harmonic mean, balancing purity against completeness. For instance, of 10,000 events flagged as large-angle about 9,820 truly are (precision 0.982), while of 10,000 genuine large-angle events about 9,800 are correctly flagged (recall 0.980). In routing terms, a precision loss lets the wrong shower type contaminate a branch, whereas a recall loss discards genuine events into the wrong expert; since both experts are trained on overlapping ranges straddling the 60° boundary and perform comparably there (Section 4.5), either error affects the final resolution only marginally. Misclassified events are concentrated near the 60° decision boundary. To quantify the impact of imperfect routing on the final angular resolution, we perform a routing ablation in which the learned classifier is replaced by an oracle (perfect) router: this changes the global Ψ_{68} by only 0.03°, indicating that residual misrouting does not dominate the reconstruction error budget. Because the boundary region (55°–60°) is where the two expert models exhibit comparable performance (Section 4.5), a misrouted event in this narrow band incurs at most a 0.06° degradation, which is small relative to the overall resolution. The SVM is therefore a reliable, autonomous routing module.

Table 1: Performance of the SVM angle classifier on the held-out test set.

Class	**Precision**	**Recall**	**F1**
Standard-angle (0°–60°)	0.988	0.986	0.987
Large-angle (60°–90°)	0.982	0.980	0.981
Overall accuracy		98.4%	

3.3. Reconstruction for Standard Incidence

For particles falling within the standard regimes (0°–60°), the shower profiles are generally complete and well-contained within the BGO calorimeter. In this regime, the official energy-weighted CoG reconstruction provides a physically robust, albeit sometimes imprecise, baseline [25, 26]. To exploit the complementary strengths of both approaches, we employ a Physics-Guided Hybrid Neural Network (PGHNN).

Input Representation: The PGHNN takes a 2-channel input tensor of shape $(14, 22, 2)$: (1) the per-event normalized energy channel $E_{ij}/\sum E$ capturing relative shower shape, and (2) the log-transformed energy channel

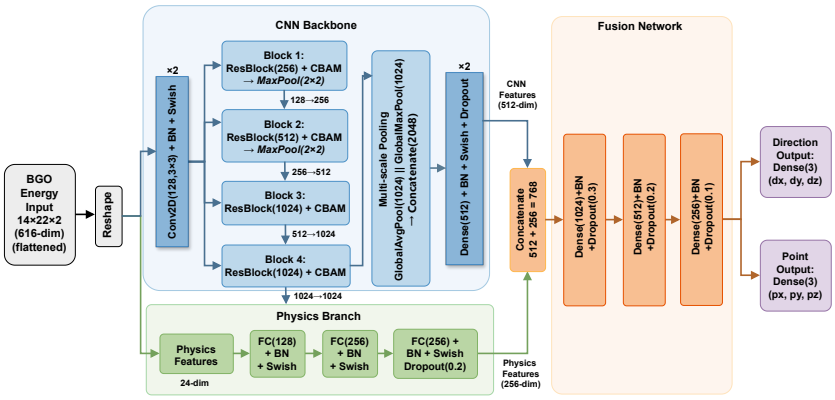


Figure 2: Schematic overview of the Physics-Guided Hybrid Neural Network (PGHNN) for standard-angle events (0°–60°). The architecture features a dual-stream structure processing BGO energy maps (2-channel input) and physics features, feeding a fusion network that drives two output heads: a direction head $\vec{D}_{\mathrm{dl}}$ (`Dense(3)`) and a point head $\vec{P}_{\mathrm{dl}}$ (`Dense(3)`). At inference the final trajectory is obtained by fusing the deep-learning prediction with the analytic (Standard BGO Algorithm) reconstruction through a gate I_{valid} derived from the validity of the analytic fit.

$\ln(1 + E_{ij})$ preserving low-energy periphery details (Section 2). This flat 616-dimensional vector is reshaped to $(14, 22, 2)$ by a dedicated `BGOInput` layer.

Architecture and Mechanism: As illustrated in Figure 2, the PGHNN has a dual-stream architecture designed to process heterogeneous data.

1. **CNN Stream:** The reshaped BGO energy map is processed by two initial Conv2D layers with 128 filters each, each followed by batch normalization and a Swish activation, and then by four residual blocks [46] equipped with Convolutional Block Attention Modules (CBAM) [47] at increasing depths (256, 512, 1024, 1024 filters); the first two blocks are followed by 2×2 max-pooling while the last two are not. Multi-scale pooling [48] (GlobalAveragePooling and GlobalMaxPooling, each over the 1024-channel feature maps) is concatenated into a 2048-dimensional vector, which is compressed through two `Dense(512)` layers with batch normalization, Swish and dropout into a 512-dimensional CNN feature vector.
2. **Physics Stream:** Simultaneously, the 24-dimensional physics feature vector is processed by a fully connected network with layers of 128, 256, and 256 neurons (each followed by batch normalization and Swish, with

the final layer additionally applying dropout of 0.2).

These two streams are concatenated (768 dimensions) and fed into a fusion network (1024–512–256 neurons, each stage with batch normalization and dropout) that produces the final representations consumed by the two output heads.

Dynamic Fusion: The PGHNN predicts two heads (Figure 2): a direction head $\vec{D}_{\text{dl}} \in \mathbb{R}^3$ and a point head $\vec{P}_{\text{dl}} \in \mathbb{R}^3$. The final direction combines the network prediction with the analytic BGO reconstruction $\vec{D}_{\text{phy}}$ (the official CoG track reconstruction of Section 1, the Standard BGO Algorithm) through a gate given by the analytic fit validity,

$$\vec{D}_{\text{final}} = I_{\text{valid}} \cdot \vec{D}_{\text{phy}} + (1 - I_{\text{valid}}) \cdot \vec{D}_{\text{dl}}, \tag{7}$$

where the gate $I_{\text{valid}} \in [0, 1]$ is a continuous validity weight that equals 1 only when the CoG track fit converges to a non-degenerate solution whose projected fit uncertainties are positive and within acceptance thresholds (and whose total energy exceeds a minimum), and decays continuously toward 0 as the fit quality degrades. This physics-derived gating guarantees a safe fallback: whenever the analytic method fails, $I_{\text{valid}} = 0$ and the prediction reduces to the deep-learning estimate $\vec{D}_{\text{dl}}$.

Physical Rationale: In the standard-angle regime the energy-weighted CoG prior remains strong, so the fusion naturally trusts the analytic reconstruction for "clean" showers in which simple geometric moments suffice, while falling back to the non-linear capacity of the network for complex showers in which the track fit degrades. This is precisely the regime in which the PGHNN is most advantageous. By contrast, at large incidence the side-leakage geometry makes $\vec{D}_{\text{phy}}$ systematically biased and frequently degenerate, so the large-angle branch (Section 3.4) does not use the analytic estimate at all: it has no analytic input and no gate, and its direction is taken directly from the deep-learning stream. The two branches therefore implement a single, physically motivated principle: trust the analytic prior where it is reliable, and fall back to learning where it is not.

Loss Function: The PGHNN is trained with a composite loss function combining mean squared error (MSE) [49], cosine similarity, and a point-regression term:

$$\mathcal{L}_{\text{standard}} = \alpha \cdot \mathcal{L}_{\text{MSE}}(\vec{D}_{\text{pred}}, \vec{D}_{\text{true}}) + \beta \cdot \mathcal{L}_{\cos}(\vec{D}_{\text{pred}}, \vec{D}_{\text{true}}) + w_{\text{pt}} \cdot \mathcal{L}_{\text{point}}, \tag{8}$$

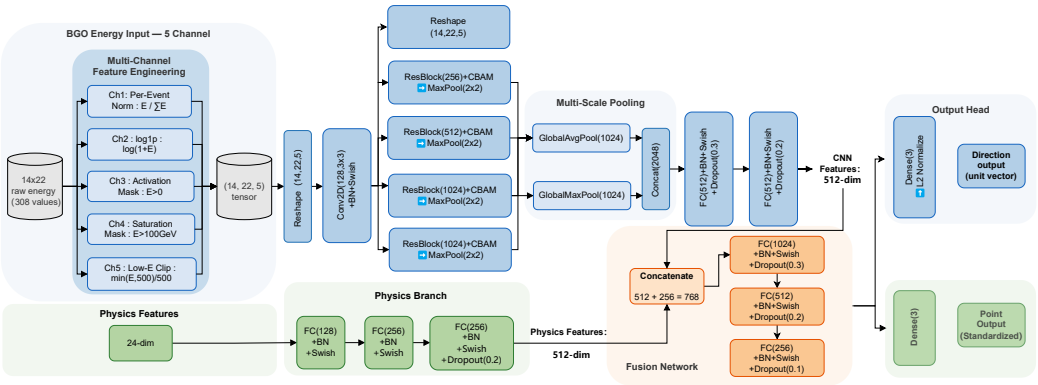


Figure 3: Schematic overview of the Deep-Learning-Dominant (DLD) architecture for large-angle events (60°–90°). The model processes a 5-channel BGO energy input (per-event normalized energy, log-transformed energy, an activation mask, a saturation mask, and a low-energy clip) together with a 24-dimensional handcrafted physics feature vector, extracts features via a CNN backbone with residual blocks [46] and CBAM attention [47], and jointly predicts direction and impact point through a multi-task fusion network.

where $\mathcal{L}_{\mathrm{MSE}} = \|\vec{D}_{\mathrm{pred}} - \vec{D}_{\mathrm{true}}\|^2$, $\mathcal{L}_{\cos} = (1 - \cos\theta)^2$, and $\mathcal{L}_{\mathrm{point}}$ is a Huber loss with threshold $\delta_H = 2.0$ applied to the predicted impact point $\vec{P}_{\mathrm{dl}}$. The coefficients are set as $\alpha = 0.1$, $\beta = 0.9$, and $w_{\mathrm{pt}} = 0.01$; the impact-point head is present, but its weight is intentionally small so that direction regression dominates training.

3.4. Reconstruction for Large Incidence

Input Representation: The DLD model consumes a 5-channel BGO energy map of shape $(14, 22, 5)$ together with a 24-dimensional handcrafted physics feature vector. The five energy channels are **(Ch1)** per-event normalized energy $E_{ij}/\sum E$, which captures the relative shower shape; **(Ch2)** log-transformed energy $\ln(1+E_{ij})$, which preserves low-energy peripheral detail and compresses the wide dynamic range of high-energy electrons; **(Ch3)** an activation mask $\mathbb{I}(E_{ij} > 0)$ indicating deposition presence; **(Ch4)** a saturation mask $\mathbb{I}(E_{ij} > 100\,\mathrm{GeV})$ flagging bars whose energy exceeds an empirical threshold, chosen because TeV-scale electromagnetic showers frequently drive single-crystal digitiser channels toward saturation; and **(Ch5)** a low-energy clip $\min(E_{ij}, 500\,\mathrm{MeV})/500$ that isolates peripheral shower topology by capping core deposits at an empirical 500 MeV ceiling. In parallel, the physics stream receives a regime-specific 24-dimensional handcrafted vector

(distinct from both the 7-dimensional morphology vector used by the Angle Classifier in Section 3.2 and the 24-dimensional standard-angle vector of Section 2); it contains, for each of the 7 layer pairs, the normalized Y-view energy, the normalized X-view energy, and the energy-weighted centroid of the Y-view bars, plus three global quantities: the logarithm of the total deposited energy, the active-bar fraction, and the index of the shower-maximum layer. These features encode geometric priors that complement the learned representations.

Network Architecture: The DLD model follows a similar dual-stream structure to the PGHNN but with important adaptations. The CNN backbone processes the 5-channel input through initial Conv2D layers (128 filters), followed by four residual blocks with Convolutional Block Attention Modules (CBAM) at increasing depths (256, 512, 1024, 1024 filters), each incorporating dropout for regularization. Multi-scale pooling combines GlobalAveragePooling and GlobalMaxPooling to capture broad spatial features. The physics stream processes the 24-dimensional feature vector through fully connected layers (128–256–256 neurons). The fusion network (1024–512–256 neurons) integrates both streams and feeds into two task-specific heads.

Multi-Task Learning Framework: The DLD architecture predicts direction and impact point jointly, in a single multi-task pass: **(1)** Direction output: a 3D unit vector representing the particle trajectory, L2-normalized to ensure $|\vec{D}| = 1$; and **(2)** Point output: the impact point coordinates on the generation sphere of radius 1.38 m (Section 2, adopted as the reference-sphere radius R), representing the true incident location from Monte Carlo simulation. The point prediction is standardized using training set statistics to ensure stable gradients.

Impact-Point Regression Weighting: Training the network to predict both direction and impact point simultaneously for large-angle events is challenging due to spatial information loss from side leakage. We therefore keep the impact-point head dormant at the start of training and anneal its loss weight only after the directional head has consolidated, following the curriculum-learning principle that harder sub-tasks should be introduced only after easier ones are mastered [50]. Concretely, a `PointWeightAnnealing` schedule sets the point-loss weight to zero for the first 15 epochs and then ramps it linearly to a final value of 0.1 over the subsequent 15 epochs, after which it is held fixed; the point loss itself is a Huber loss with $\delta = 0.5$. This delayed, gradual introduction prevents the point-head gradient from disrupting the directional head during early training and avoids convergence to local

minima caused by ambiguous impact points in leaking showers.

Loss Function Design: The DLD model employs a specialized loss function accounting for the specific challenges of large-angle reconstruction. Table 2 summarizes the key differences between the standard and large-angle models.

Table 2: Comparison of loss function components between the PGHNN (standard-angle) and DLD (large-angle) models.

Component	**Standard** (0°–60°)	**Large-Angle** (60°–90°)
Direction Loss	$0.1\mathcal{L}_{\text{MSE}} + 0.9(1-\cos\theta)^2$	$0.1\mathcal{L}_{\text{MSE}} + 0.9(1-\cos\theta) + 0.05\mathcal{L}_{\text{flip}}$
Flip Penalty	Not used	$\lambda\,\mathbb{I}(z_{\text{p}}z_{\text{t}}<0)$, λ=0.05
Point Loss	$0.01\,\mathcal{L}_{\text{Huber},\delta=2.0}$ (minimal weight)	$w_{\text{pt}}(t)\,\mathcal{L}_{\text{Huber},\delta=0.5}$, annealed 0→0.1 (ep. 15–30)

The direction loss combines MSE ($\alpha = 0.10$), the linear cosine term $1 - \cos\theta$ ($\beta = 0.90$), and a flip penalty ($\lambda = 0.05$) that penalizes 180° directional reversals when z_{pred} and z_{true} have opposite signs. The impact-point head uses a Huber loss with $\delta = 0.5$ whose weight $w_{\text{pt}}(t)$ is annealed from 0 to 0.1 over training epochs 15–30.

Hemisphere ambiguity. A BGO-only reconstruction cannot, in principle, determine a priori whether a shower was initiated by a downward-going particle entering from above the detector or an upward-going particle entering from below, because the calorimeter records only energy deposition and not the absolute direction of incidence. The flip penalty above resolves this at training time by enforcing consistency with the Monte Carlo truth: it penalizes predictions whose z-component has the opposite sign to the true direction ($z_{\text{pred}} \cdot z_{\text{true}} < 0$), so the network learns to orient the predicted vector according to the true incident hemisphere of the training sample. Because the training data in this work are downward-going events ($0° \leq \theta \leq 90°$), the model implicitly adopts the downward convention, and at inference it assumes downward incidence, a valid assumption for the validated 0°–90° range. Consequently, any residual hemisphere mis-assignment is expected to be negligible for the validated downward-going 0°–90° sample; correctly resolving upward-going events in the future 90°–130° extension will require either an explicit hemisphere classifier or the use of the earliest shower-development layer as a directional prior.

3.5. Training Protocol and Implementation Details

For reproducibility we summarize the training configuration of the two expert models; both share the backbone described in Sections 3.3 and 3.4 and are trained independently on overlapping but not identical angular subranges.

Optimizer and learning-rate schedule. The two models use different schedules. The standard-angle model is optimized with Adam using a `CyclicalLearningRate` schedule (base_lr $= 10^{-5}$, max_lr $= 8 \times 10^{-4}$, step_size $= 2500$; clipnorm $= 1.0$; default $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-7}$). The large-angle model is optimized with Adam (clipnorm $= 1.0$) using a `CyclicalLearningRate` schedule (base_lr $= 10^{-5}$, max_lr $= 8 \times 10^{-4}$, step_size $= 2500$) preceded by a linear warm-up of 3000 steps from an initial rate of 2×10^{-5} (`WarmupLR`).

Batch size and epochs. Both models are trained with batch size 2048 for up to 200 epochs each (with early stopping, patience 20, monitoring the validation angular error; training typically converges well before the nominal maximum).

Losses. The composite losses are those defined in Sections 3.3 and 3.4. The standard-angle model combines an MSE term ($\alpha = 0.1$), the squared cosine term $(1 - \cos\theta)^2$ ($\beta = 0.9$), and a small impact-point Huber term ($w_{\mathrm{pt}} = 0.01$, $\mathcal{L}_{\mathrm{Huber},\delta=2.0}$); it *does* output an impact-point head, whose weight is intentionally negligible so that direction regression dominates. The large-angle model additionally includes a flip penalty and an impact-point Huber loss ($\delta = 0.5$) whose weight is annealed from 0 to 0.1 over epochs 15–30.

Data scale and split. Training uses the Monte Carlo samples of Section 2: of the 11,894,154 events spanning 100 GeV–1 TeV and zenith range 0°–90°, the 80% training partition (9,515,323 events) is used for model training and the remaining 20% for evaluation, stratified by (θ, E).

Training angular support. The standard-angle PGHNN is trained on events in 0°–70° and the large-angle DLD on events in 50°–90°, so both models see showers from either side of the 60° routing threshold. The SVM still routes events at 60°; extending the training ranges does not change this rule.

Hardware and inference cost. All models were trained on a single NVIDIA A100 (80 GB) GPU with CUDA 11.7, cuDNN 8.6.0.163, TensorFlow 2.10, and Python 3.10. A single event is reconstructed in approximately 1.5 ms on the same GPU, and a full training run requires approximately

20 h; these figures confirm that the method is feasible for offline ground processing of the DAMPE downlink stream.

4. Performance Evaluation

4.1. Evaluation Metrics

The quality of the trajectory reconstruction is quantified using three primary metrics: **Angular Resolution (Ψ_{68}), Cosine Similarity, and Reconstruction Efficiency.**

Angular Resolution (Ψ_{68}) is defined as the space angle containing 68% of the reconstructed events relative to the true MC truth direction. The space angle error ψ for a single event is calculated as:

$$\psi = \arccos(\hat{\mathbf{v}}_{\text{pred}} \cdot \hat{\mathbf{v}}_{\text{true}}) \tag{9}$$

where $\hat{\mathbf{v}}_{\text{pred}}$ and $\hat{\mathbf{v}}_{\text{true}}$ are the unit vectors of the predicted and true particle directions, respectively. The metric Ψ_{68} represents the 68th percentile of the distribution of ψ.

Cosine Similarity measures the alignment accuracy between the predicted and true vectors, providing a direct metric for directional coherence:

$$\text{Cosine Similarity} = \frac{\mathbf{v}_{\text{pred}} \cdot \mathbf{v}_{\text{true}}}{\|\mathbf{v}_{\text{pred}}\|\|\mathbf{v}_{\text{true}}\|} \tag{10}$$

Values closer to 1.0 indicate higher reconstruction accuracy.

Reconstruction Efficiency evaluates the robustness of the algorithm. It is defined as the fraction of the events entering the evaluation that are reconstructed with an angular error below a tolerance threshold of 10°:

$$\epsilon = \frac{N(\psi < 10^\circ)}{N_{\text{total}}} \tag{11}$$

where N_{total} is the number of simulated events in the evaluation sample. The same denominator is used for every method compared in this work, so that the efficiencies are directly comparable, and no trigger-efficiency correction is applied. Events with $\psi \geq 10^\circ$ are counted as reconstruction failures.

In addition to the directional metrics above, we introduce the Arc-Length Error to evaluate the impact point reconstruction precision of the two expert models. Both models predict the incident point at the generation sphere of Section 2 (radius 1.38 m), so the reconstructed trajectory is the straight

line that passes through the predicted impact point $\vec{P}_{\text{pred}}$ along the predicted direction $\hat{\mathbf{v}}_{\text{pred}}$. The Arc-Length Error is defined as the great-circle distance on the reference sphere between the true incident point and the upstream (entrance-side) intersection of this line with the same sphere. Let $\hat{\mathbf{u}}_{\text{true}}$ and $\hat{\mathbf{u}}_{\text{pred}}$ denote the unit vectors from the sphere center to the true point and to that intersection, respectively. The arc-length error s is given by:

$$s = R \cdot \arccos(\hat{\mathbf{u}}_{\text{true}} \cdot \hat{\mathbf{u}}_{\text{pred}}) \tag{12}$$

where R is the reference-sphere radius, taken to be the generation radius of 1.38 m. The vector $\hat{\mathbf{u}}_{\text{pred}}$ therefore points to where the reconstructed line crosses the entrance surface, which is the incident position a downstream analysis would use. Because the predicted impact point is confined to a thin shell around the reference sphere, this intersection is dominated by the tangential position of $\vec{P}_{\text{pred}}$ and is only weakly sensitive to $\hat{\mathbf{v}}_{\text{pred}}$ (Section 4); s is thus a measure of the entry-point displacement, that is, of the impact-point prediction, and not a restatement of the angular resolution.

4.2. Reconstruction Efficiency and Accuracy

A major limitation of the Standard BGO Algorithm is the loss of usable events at large incidence angles. Table 3 shows that its efficiency, defined by the $\psi < 10^\circ$ criterion of Section 4, is 98.73% in 0°–30° and 91.06% in 30°–60°, but falls to 38.58% in 60°–90°. Combined with the angular resolution of $\Psi_{68} = 32.73^\circ$ in that band (Table 4), this means that the majority of large-angle events are still reconstructed with unusable directions. The degradation follows from the failure of the shower axis fit when the shower maximum leaks out of the calorimeter. In contrast, the PGHNN and DLD framework achieves near 100% efficiency across the entire 0°–90° domain, thereby extending the detector field of view.

Table 3: Comparison of Reconstruction Efficiency across different angular ranges.

Method	**0°–30°**	**30°–60°**	**60°–90°**
Standard BGO Algorithm	98.73%	91.06%	38.58%
PGHNN (This work)	**100%**	**99.47%**	—-
DLD (This work)	—-	—-	**99.50%**

Table 4 compares the angular resolution (Ψ_{68}) of the methods. In the standard-angle region (0°–60°), the PGHNN reduces Ψ_{68} from 3.78° to 1.40°

(a factor of ≈ 2.7) by combining physical priors with the dynamic fusion mechanism. In the large-angle region (60°–90°), the Standard BGO Algorithm degrades to $\Psi_{68} = 32.73°$ (with a reconstruction efficiency of 38.58% under the $\psi < 10°$ criterion of Section 4), whereas the DLD maintains 1.31° (a factor of ≈ 25).

Table 4: Comparison of Angular Resolution (Ψ_{68}, lower is better). The Standard BGO Algorithm entries are obtained with the official DAMPE energy-weighted CoG track reconstruction applied to the same Monte Carlo electron sample as the two deep-learning models. The conventional BGO+STK pipeline is listed for reference only: quoted at 10 GeV from Ref. [12], it operates within 0°–60° and is not evaluated on the 100 GeV–1 TeV sample used here; it is therefore not directly comparable to the BGO-only methods in the same energy band.

Method	**68% Quantile (degree)**
Conventional BGO+STK pipeline (0°–60°)	**0.2**
Standard BGO Algorithm (0°–60°)	3.78
Standard BGO Algorithm (60°–90°)	32.73
PGHNN Method (0°–60°)	1.40
DLD Method (60°–90°)	**1.31**

Figure 4 presents the direction residual distributions for both angular regimes. In the standard-angle regime (top panel), the zenith and azimuth residuals exhibit approximately symmetric, unimodal distributions centered near zero, with the space angle error distribution peaking at approximately 0.8°. The large-angle regime (bottom panel) shows a similar overall shape but with a modestly broader width, particularly in the azimuth residual and space angle error, reflecting the increased reconstruction challenge posed by shower leakage. Both distributions are centered near zero with negligible mean offset, confirming the high accuracy of the reconstructed directions.

4.3. Energy Dependence and Stability

The performance stability across primary particle energies is evaluated in Figure 5. The Standard BGO Algorithm (black dashed line), evaluated on the same Monte Carlo sample in the 0°–60° range, shows a mild energy dependence: Ψ_{68} is approximately 4.0° at 100 GeV, rises slightly to $\sim 4.05°$ around 150 GeV, and then decreases steadily to $\sim 3.5°$ towards 1 TeV. The energy-integrated value of 3.78° quoted in Table 4 is consistent with these per-bin values. Such behaviour is characteristic of conventional calorimetric

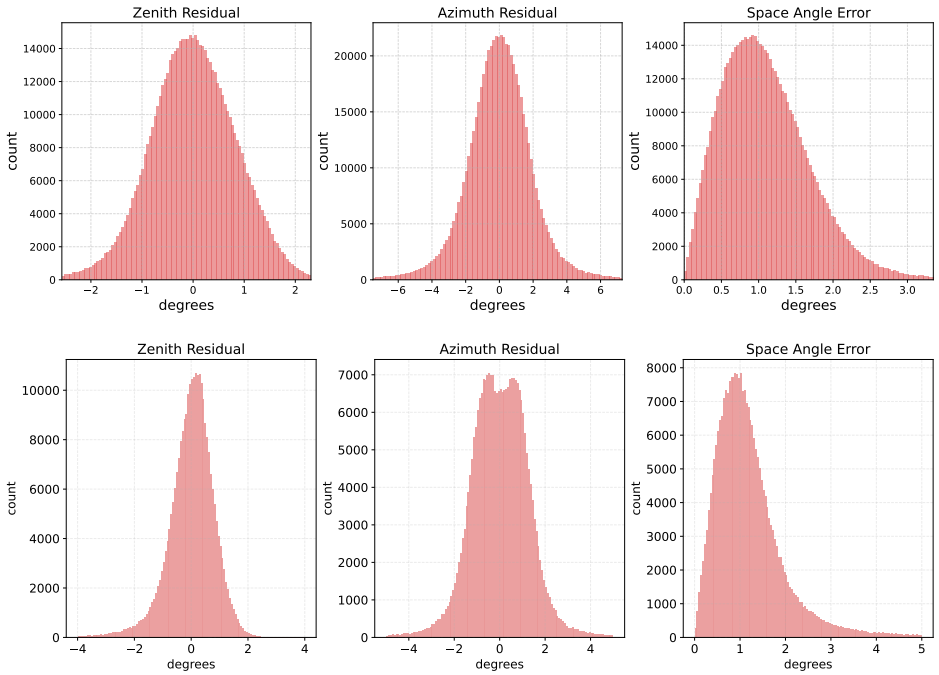


Figure 4: Direction residual distributions for the standard-angle regime (top, PGHNN) and the large-angle regime (bottom, DLD). Within each panel (left to right): zenith residual, azimuth residual, and space angle error.

methods, whose precision is limited by the statistics of the energy deposition. In contrast, the PGHNN (red) and DLD (blue) curves change much less with energy than the standard algorithm, improving slightly from $\sim 1.5°$ and $\sim 1.4°$ to $\sim 1.3°$ and $\sim 1.2°$, respectively. This mild energy response suggests that the physics-guided PGHNN extracts topological features of the shower axis that are less sensitive to energy scale, compensating for the fluctuations that limit analytical approaches at lower energies.

For the large-angle regime (60°–90°), the DLD model exhibits a systematic energy dependence, as shown by the blue curve in Figure 5. Across the 100 GeV to ∼900 GeV range, the angular resolution improves monotonically from approximately 1.4° to 1.2°. While this energy dependence is more pronounced than that of the PGHNN in the standard-angle regime, the absolute resolution remains well below 1.5° across the entire range. This behavior indicates that the five-channel input representation and the curriculum learning strategy enable the network to extract directional features even when a substantial fraction of the shower energy escapes the calorimeter.

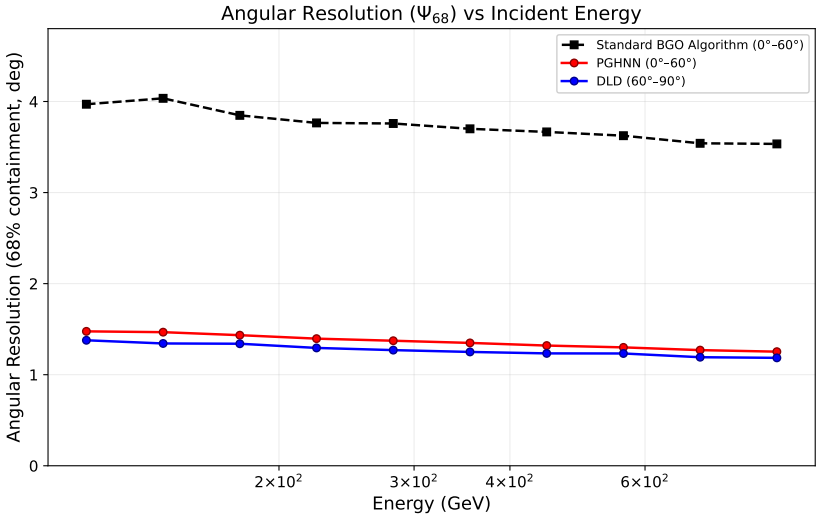


Figure 5: Angular resolution (Ψ_{68}) versus incident energy over the 100 GeV–1 TeV range, comparing the Standard BGO Algorithm (black dashed, 0°–60°), the PGHNN (0°–60°, red) and the DLD (60°–90°, blue). The energy-integrated Ψ_{68} of the Standard BGO Algorithm is 3.78° (Table 4).

4.4. Impact Point Reconstruction Precision

In addition to directional accuracy, the joint prediction of the incident point by the DLD model enables a direct assessment of impact point reconstruction precision via the Arc-Length Error metric defined in Section 4. Figure 6 presents the distributions of arc-length errors for both angular regimes overlaid as normalized probability densities. In the standard-angle range (0°–60°), the arc-length error achieves a P68 of 32.99 mm and a P95 of 55.17 mm, corresponding to an angular deviation of approximately 1.37° and 2.29° on the reference sphere, respectively. For the large-angle regime (60°–90°), where side leakage complicates both direction and impact point estimation, the arc-length error achieves a P68 of 38.48 mm and a P95 of 80.99 mm, equivalent to angular deviations of approximately 1.60° and 3.36°, respectively. The two distributions differ chiefly in their tail behaviour: the standard-angle distribution decays rapidly beyond $\sim$ 55 mm, whereas the large-angle distribution exhibits a noticeably heavier tail extending beyond $\sim$ 100 mm, reflecting the increased challenge of impact-point reconstruction under grazing-incidence geometry.

To investigate the angular dependence of the impact point precision, Fig-

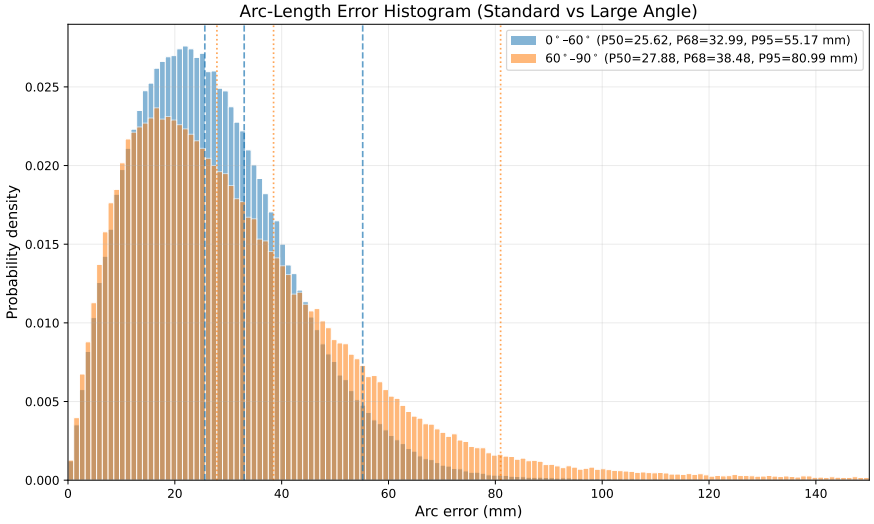
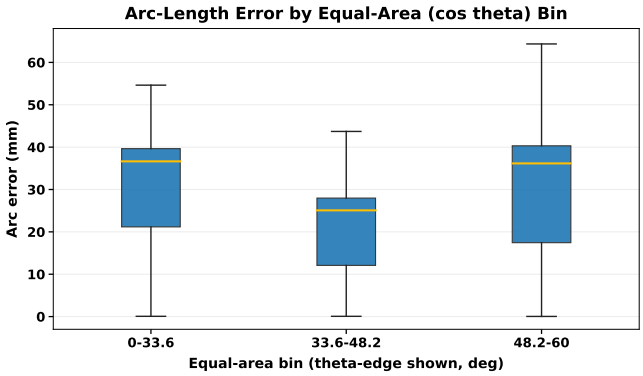


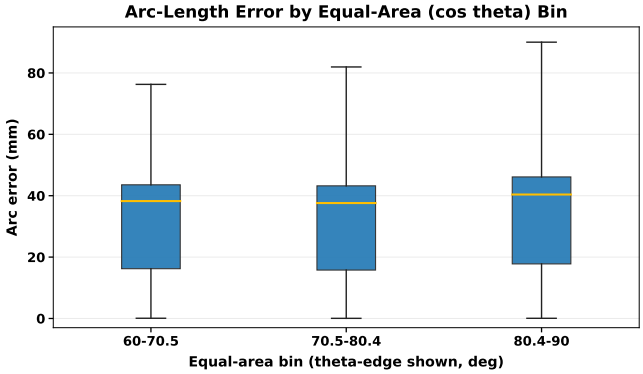
Figure 6: Arc-length error distributions for the standard-angle regime (blue, 0°–60°) and the large-angle regime (orange, 60°–90°), overlaid as normalized probability densities. Vertical lines in same color mark the P50, P68 and P95 percentiles of each distribution.

ure 7 shows the arc-length error statistics binned by equal-area intervals in $\cos\theta$. In the standard-angle regime, the P68 values are approximately 37 mm for the 0°–33.6° bin, 25 mm for the 33.6°–48.2° bin, and 36 mm for the 48.2°–60° bin. In the large-angle regime, the P68 values are approximately 38 mm for the 60°–70.5° bin, 38 mm for the 70.5°–80.4° bin, and 41 mm for the 80.4°–90° bin. The P95 values remain below 100 mm across all bins, confirming that the multi-task learning framework maintains acceptable impact point precision even under extreme leakage conditions.

The arc-length error is not a second measurement of the direction: it records where the reconstructed line crosses the entrance sphere. The predicted impact point lies close to that sphere, so the reconstructed line meets it close to $\vec{P}_{\text{pred}}$; only the residual radial offset of the prediction acts as a lever arm for the directional error, and for the resolutions reported here ($\sim 1.3°$–$1.4°$) it contributes well under a millimetre. The arc-length error is therefore a measure of the tangential accuracy of the impact-point prediction, and the two regimes can be compared directly: the entry point is located to 3.3 cm (P68) at standard angles and 3.8 cm at large angles, a degradation of only about 17% where the shower is truncated by side leakage. The ability to reconstruct the incident point with centimeter-level precision (P68 $<$ 5 cm)



(a) Standard-angle regime (0°–60°).



(b) Large-angle regime (60°–90°).

Figure 7: Arc-length error statistics by equal-area $\cos\theta$ bins for (a) the standard-angle regime and (b) the large-angle regime. The box spans Q1–Q3, the yellow line marks P68, and the whiskers extend to the minimum and P95.

is scientifically valuable, as it enables the extension of particle identification algorithms (particularly electron–hadron discrimination) into the large-angle regime [2], where the shower morphology relative to the precise axis is a relevant input to rejection variables.

4.5. Boundary Analysis and Transition Strategy

A hierarchical pipeline that routes events with a hard threshold at 60° is intrinsically sensitive to how the two expert models behave at that boundary. If each expert is trained only on its own, disjoint angular domain, neither network has seen showers from the other side of the threshold; events close to

60° are then reconstructed by a model operating at the edge of its training support, and the routed angular resolution can change abruptly when the classifier switches branches. We therefore examine the continuity of Ψ_{68} across the routing threshold and describe a training-domain overlap that removes this discontinuity.

The routing rule is unchanged: the SVM classifier of Section 3.2 assigns 0°–60° events to the PGHNN and 60°–90° events to the DLD, and inference remains a single hard routing decision. We extend only the angular support of the training samples. The standard-angle model is trained on events spanning 0°–70° instead of 0°–60°, and the large-angle model on 50°–90° instead of 60°–90° (Section 3.5). Each expert thus sees a 10° overlap band straddling the nominal boundary, while the overall Monte Carlo sample of Section 2 is unchanged.

Figure 8 shows the resulting stitched Ψ_{68} curve over 0°–90°, where each branch is evaluated only in the angular interval to which it is routed. The two branches now join continuously at 60°: both give $\Psi_{68} \approx 1.2°$–$1.3°$ at the boundary, and the step present when the models were trained on disjoint domains is no longer visible. The resolution stays below $\sim 1.7°$ over the full range: it rises from $\sim 0.8°$ at small zenith angles to $\sim 1.5°$ near 35°, passes through a shallow local minimum of $\sim 1.1°$ near 63°, and shows a broad local maximum of $\sim 1.7°$ near 79° before falling back to $\sim 1.3°$ at 90°. No extrapolation beyond the training support of either model is shown in this figure.

The choice of 60° rests on the two mechanisms identified in Section 3.1—the STK fiducial-volume cutoff and the degradation of the Standard BGO Algorithm, which coincide at this angle—together with the continuity of the routed resolution at that angle. The overlap band regularizes the boundary during training: both experts have been trained on showers from either side of the threshold, so the residual misclassifications of the SVM, which are concentrated near 60° (Section 3.2), no longer produce a localized degradation in resolution. The two predictions are not blended at inference time. A continuous soft-transition scheme, in which the two predictions are interpolated across the overlap band, is left for future work.

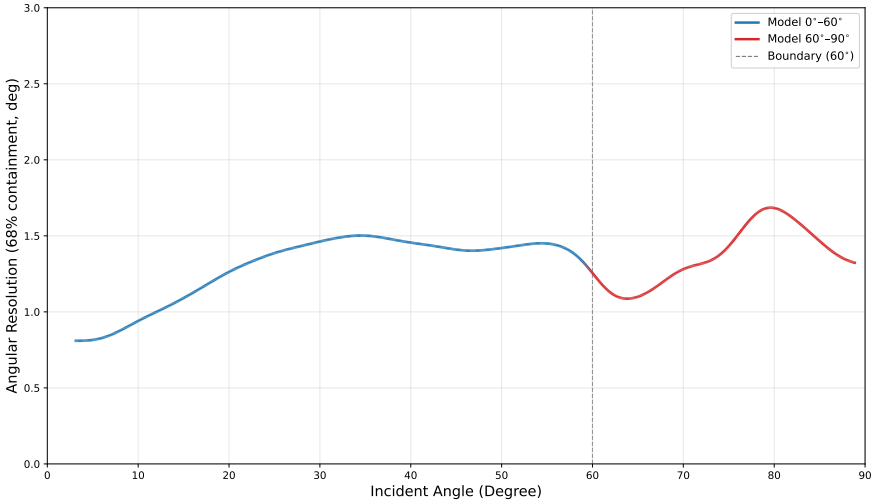


Figure 8: Routed angular resolution (Ψ_{68}) versus incident angle across 0°–90° for the PGHNN (blue, labelled "Model 0°–60°") and the DLD (red, labelled "Model 60°–90°"), each evaluated in the interval to which it is routed. Both models were trained on overlapping angular ranges (0°–70° and 50°–90°), so the two branches join continuously at the routing threshold. The vertical dashed line marks the 60° boundary.

4.6. Ablation Study

To substantiate the claim that the physics-guided components, rather than the deep-learning capacity alone, drive the reported accuracy, we perform an ablation study on the standard-angle (PGHNN) model, measuring the angular resolution Ψ_{68} under each modification. The results are summarized in Table 5.

Table 5: Ablation study on the standard-angle PGHNN model. Ψ_{68} (degree, lower is better) is reported for the full model and for each modification.

Configuration	Ψ_{68} **(deg)**
Full PGHNN (24 physics features + dynamic fusion)	1.40
– without the 24 physics features (energy-map only)	1.81
– without dynamic fusion (always trust DL, $I_{\text{valid}} \equiv 0$)	1.56
– reduced input (single channel)	1.49

Removing the hand-crafted physics features degrades the resolution by 0.41° (from 1.40° to 1.81°), confirming that the feature engineering supplies

geometric priors that the convolutional backbone alone exploits only partially. Disabling the physics-guided dynamic fusion (forcing the prediction to the deep-learning stream) yields a 0.16° change (1.56° vs 1.40°), indicating that the arbitration between the analytical and data-driven branches contributes modestly but consistently. Reducing the input to a single channel provides a 0.09° degradation (1.49° vs 1.40°), confirming the value of the dual-channel representation. Collectively, these ablations demonstrate that the improvements arise from concrete, separable components (physics-informed feature engineering, dynamic physics-guided fusion, and dual-channel input) within a measurable "gray-box" framework, rather than from an opaque black-box model; each contribution is measured, not merely asserted.

5. Discussion

5.1. Application to Flight Data

The transition from Monte Carlo simulations to on-orbit flight data is complicated by the absence of ground-truth incident-angle labels. The SVM classifier of Section 3.2 addresses this by routing events from shower morphology alone. As a first check on real detector data, we apply the resulting pipeline to a sample of flight electron candidates that leave a reconstructable track in both the STK and the BGO calorimeter, with STK-derived zenith angles between 30° and 71° ($N = 391{,}148$). We use the STK track as an independent reference direction and define the angular difference ψ as the space angle between the BGO-only prediction and that track, using the definition of Section 4 with the STK track in place of the Monte Carlo truth.

Figure 9 compares the two zenith distributions and their cumulative distributions, together with their bin-by-bin ratio. The shapes agree closely over most of the 30°–71° interval and peak between 45° and 52°; the ratio remains within 0.95–1.13, with a median absolute deviation from unity of 0.04. The largest local differences lie in $\sim$ 53°–62°, where the ratio reaches 1.13; smaller but systematic deficits appear at 30°–40° ($\sim$ 0.96) and 45°–50° ($\sim$ 0.97). These excursions exceed the bin-wise statistical uncertainty of $\sim$ 0.01–0.02. Both samples show the dip near 54°–55°, which we attribute to the event selection; the model does not introduce it. Towards 71° the bins become sparsely populated, the ratio uncertainty grows to $\sim$ 0.15, and bins with fewer than 50 events in either sample are omitted from the ratio panel. Their origin is not established by the present cross-check. Figure 10 shows the distribution of ψ: it is unimodal, with a maximum near 1° and a long

tail out to 8°, and gives a median of 1.44°, a 68th percentile of 1.98° and a 95th percentile of 5.56°.

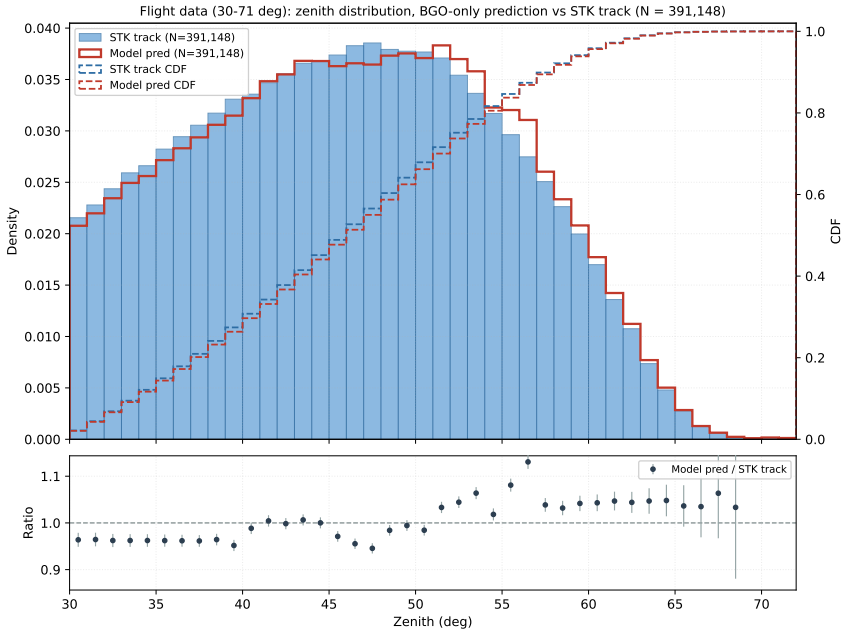


Figure 9: Comparison of the zenith-angle distributions reconstructed by the BGO-only model and by the STK for flight electron candidates with both detectors traversed ($30° \leq \theta \leq 71°$, $N = 391{,}148$). *Top*: filled blue histogram, STK track; red step line, model prediction ("Model pred" in the legend), both normalized to unit area; the dashed blue and red curves are the corresponding cumulative distributions on the right-hand axis. *Bottom*: ratio of the model prediction to the STK track in each 1° zenith bin. The error bars give the statistical uncertainty $\sigma(R)/R = \sqrt{1/N_{\mathrm{pred}} + 1/N_{\mathrm{STK}}}$; the dashed horizontal line marks unity. Only bins with at least 50 events in both samples are shown, and the two panels share the horizontal axis.

Figure 11 resolves the same quantity in logarithmically equal-width bins of the BGO reconstructed energy between 83.3 and 941.2 GeV. The median falls from $\sim 1.6°$ in the lowest bin to $\sim 1.2°$ in the highest, and the 68th percentile from $\sim 2.2°$ to $\sim 1.6°$. The 95th percentile declines more gently, from $\sim 5.9°$ to $\sim 5.1°$, and shows bin-to-bin scatter of up to $\sim 0.8°$, while the number of entries per bin drops from $\sim 1.2 \times 10^5$ to a few thousand. The median and the 68th percentile describe the bulk of the events and change smoothly with energy; the 95th percentile is set by a small subpopulation of

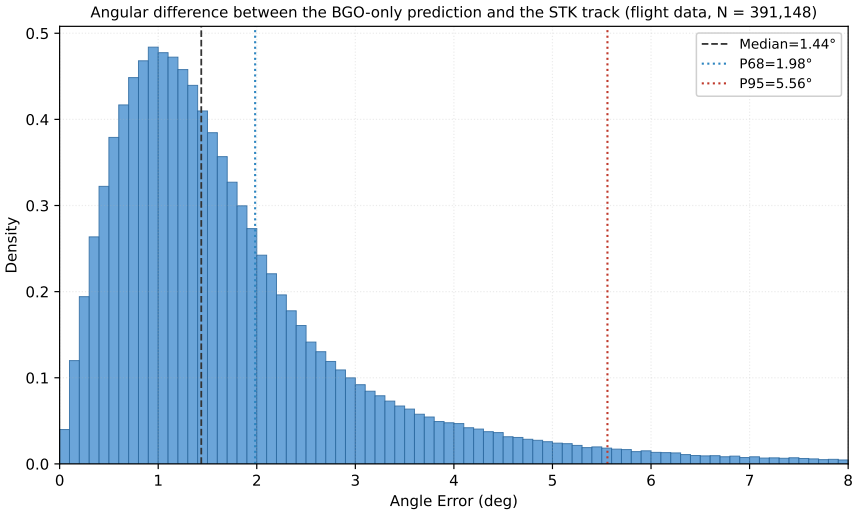


Figure 10: Distribution of the space-angle difference ψ between the BGO-only model prediction and the STK track for the flight sample of Figure 9. The dashed, dotted and dash-dotted vertical lines mark the median (1.44°), the 68th percentile (1.98°) and the 95th percentile (5.56°), respectively. This quantity describes the agreement between two reconstructions. The STK track is not a truth direction, so ψ is not an absolute angular resolution.

poorly reconstructed events (events misrouted near the 60° boundary, events with degraded STK track quality at large zenith angles, and residual contamination), whose fraction can differ from bin to bin. Fewer events widen the statistical uncertainty of this tail quantile in the highest bins, but the largest excursions appear in bins with ample statistics, so the finite bin population does not by itself explain the scatter. We do not assign a single cause to it in the present analysis.

The STK track has a finite angular resolution of its own and is not a truth label. We use ψ to compare the BGO-only prediction with that track, so the values reported above describe how well two reconstructions agree; the absolute angular resolution of the BGO-only model cannot be determined from this sample. The flight sample contains only events with an STK track, and its energy spectrum and zenith distribution differ from the isotropic, power-law Monte Carlo sample of Section 2. The 68th percentile of 1.98° is measured on this sample against the STK track and is not directly comparable to the Monte Carlo Ψ_{68} of 1.40° and 1.31°. A difference between the flight and Monte Carlo values would not, on its own, indicate a degradation or confirm

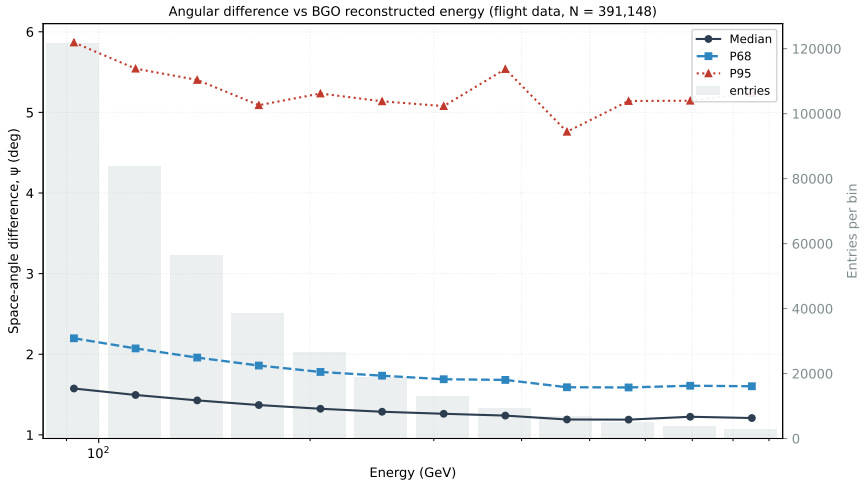


Figure 11: Energy dependence of the space-angle difference between the BGO-only prediction and the STK track ($N = 391{,}148$), binned in logarithmically equal-width intervals of the BGO reconstructed energy between 83.3 and 941.2 GeV. Black circles: median; blue squares: 68th percentile; red triangles: 95th percentile (left axis). Grey bars show the number of events per energy bin (right axis). The 95th percentile declines mildly with energy and shows bin-to-bin scatter on top of that trend. As a tail statistic, it responds more strongly than the median or the 68th percentile to the small subpopulation of poorly reconstructed events.

the simulation. The comparison shows degree-level agreement between the BGO-only predictions and an independent tracker-based direction, and Figure 9 shows no gross distortion of the zenith distribution. The energy trend in Figure 11 resembles that of Figure 5. The sample does not extend into the 71°–90° region where the STK is blind and which motivates the large-angle branch. The 60°–71° events that retain an STK track form a biased subset of the large-angle population. The dedicated study of detector effects (dead channels, pedestal and electronic noise, gain drift and pile-up) and the full data/MC closure test on an unbiased control sample remain ongoing. We expect the hemisphere assumption of Section 3.4 to hold over the validated 0°–90° range, and we will revisit it when extending the reconstruction to upward-going events.

5.2. Physics Impact

The primary scientific motivation for developing this full-angle reconstruction framework is to maximize the scientific yield of the DAMPE mission

by expanding its effective observation window.

Acceptance Enhancement: The standard reconstruction pipeline, constrained by the geometric requirements of the STK and PSD, effectively utilizes only a limited fraction of the potential field of view of the BGO calorimeter. To quantify the gain from extending reliable BGO-only reconstruction from the conventional 60° operating range to the full 90° hemisphere, we compute the *geometric acceptance* of the DAMPE BGO calorimeter for an isotropic cosmic-ray flux. Unlike a ground-based experiment, in which a flat detector module views the sky through a $\cos\theta$ projected-area weighting, DAMPE is a free-flying, six-faced box: a particle can enter through the top face *or* any of the four side faces, so the acceptance must be evaluated from the full projected area of the box rather than from a single top-face projection. For a rectangular box of dimensions $L_x \times L_y \times L_z$, the Cauchy projection formula gives the projected area along a direction (θ, ϕ) as

$$A_{\rm proj}(\theta,\phi) = L_x L_y \cos\theta + L_x L_z |\sin\theta \sin\phi| + L_y L_z |\sin\theta \cos\phi|, \tag{13}$$

where the first term is the (top) face, visible for $\theta < 90°$, and the remaining two terms are the four side faces. The geometric acceptance within zenith angle $\theta_{\rm max}$ is the integral of this projected area over the solid angle,

$$\begin{aligned} G(\theta_{\rm max}) &= \int_0^{2\pi}\int_0^{\theta_{\rm max}} A_{\rm proj}(\theta,\phi)\, \sin\theta\, d\theta\, d\phi \\ &= \pi L_x L_y \sin^2\theta_{\rm max} + 2L_z(L_x+L_y)\big(\theta_{\rm max} - \sin\theta_{\rm max}\cos\theta_{\rm max}\big). \end{aligned} \tag{14}$$

The DAMPE BGO calorimeter comprises 308 BGO bars arranged in 14 layers of 22 bars in orthogonally alternating orientation, each bar measuring $2.5 \times 2.5 \times 60$ cm^3. Its active bounding box therefore has transverse dimensions $L_x = 60$ cm (bar length) and $L_y = 22 \times 2.5 = 55$ cm (packed width of one layer), and depth $L_z = 14 \times 2.5 = 35$ cm (corresponding to ~ 31 radiation lengths). Adopting the conventional reliable limit $\theta_c \approx 60°$, beyond which the STK fiducial volume is progressively bypassed (consistent with the in-flight performance of the DAMPE silicon tracker [12]), the acceptance ratio to the full 90° hemisphere is

$$\frac{G(90°)}{G(60°)} = \frac{\pi L_x L_y + 2L_z(L_x+L_y)\,\pi/2}{\pi L_x L_y \sin^2 60° + 2L_z(L_x+L_y)\big(\pi/3 - \sin 60°\cos 60°\big)} \approx 1.81. \tag{15}$$

The quantity that a physics analysis actually gains is the *effective* acceptance, i.e. the geometric acceptance weighted by the probability that an event is reconstructed with a usable direction. Equation (15) is the ratio of effective acceptances under the assumption that the conventional BGO+STK chain is fully efficient below θ_c and contributes nothing above it, while the new chain is fully efficient everywhere. The second assumption is borne out by the measured efficiencies of 99.47% in 30°–60° and 99.50% in 60°–90° (Table 3). We therefore quote ≈ 1.8 as the headline acceptance gain, understood as an effective-acceptance ratio rather than a purely geometric one. Equivalently, the newly accessible 60°–90° annulus alone contributes a geometric acceptance equal to $\sim 80\%$ of the previously covered 0°–60° range. This $\sim 1.8\times$ increase in statistics is valuable for studies of rare events, such as the search for monochromatic gamma-ray lines or the precise measurement of the cosmic ray electron spectrum at TeV energies, where the flux is extremely low and statistical uncertainty is the dominant error source.

Background Rejection: The improved trajectory reconstruction also has implications for particle identification, specifically electron–proton discrimination, which relies heavily on analyzing the three-dimensional shower morphology relative to the shower axis. A precise axis reconstruction allows for a more accurate definition of the "shower core" and "isolation cone," which are central to the calculation of rejection variables. By providing a reliable track even for large-angle events, the method extends the domain over which electron–proton separation algorithms can operate, opening the high-inclination regime to the same discrimination tools already used near normal incidence. We note that this is a forward-looking capability: a dedicated quantitative rejection study on a mixed proton–electron Monte Carlo sample, including the definition of appropriate shower-core and isolation-cone variables in the large-angle regime, has not yet been performed and is deferred to future work.

5.3. Systematic Uncertainties and Monte Carlo Limitations

The resolution and efficiency values reported above come from GEANT4 simulation. The flight-data comparison of Section 5.1 checks the reconstructed directions against STK tracks, which are not truth labels, so it does not replace a data/MC closure test. Real detector conditions add effects that the nominal simulation omits: dead or noisy channels, pedestal and electronic noise, gain drift over the mission lifetime, and pile-up at high

rates. We inject representative detector effects into the simulation and re-evaluate the angular resolution and efficiency. Under the injected perturbations the standard-angle resolution degrades from 1.40° to 1.52°, and the large-angle resolution from 1.31° to 1.66°; the efficiency stays above 98.0% in both regimes. These preliminary numbers suggest that the reconstruction tolerates realistic detector perturbations. A full data/MC closure test on an unbiased flight-data control sample remains the definitive validation and is ongoing. Until that comparison is available, the absolute resolution figures should be read as Monte Carlo projections. The dependence of the shower model on the Geant4 physics list has not been explored systematically.

5.4. Limitations and Uncertainty Quantification

Scope limitations. The present validation is restricted to electron primaries in the 100 GeV–1 TeV energy range and the 0°–90° zenith band. The flight-data cross-check covers only events with an STK track in the 30°–71° range, and it leaves the STK-blind region above 71° untested. The scientific motivation (TeV electron spectra, gamma-ray lines) extends to higher energies, and the complementary proton, helium, and heavier-nuclei channels, as well as the upward-going 90°–130° regime, are explicitly deferred to future work. Consequently, the acceptance gain and resolution figures should not be extrapolated beyond the validated electron / 100 GeV–1 TeV / 0°–90° domain without further study.

Input-engineering choices. The thresholds used in the large-angle five-channel input (100 GeV saturation flag and 500 MeV low-energy clip) are empirical engineering choices that were found to improve reconstruction quality on the 100 GeV–1 TeV electron sample. They are not derived from first-principles detector calibration, and their optimal values may differ for other primary species or energy regimes. A systematic threshold scan is outside the scope of the present work and remains useful future work.

Direction uncertainty quantification. The reported Ψ_{68} is an empirical 68% containment radius of the space-angle error distribution; across the energy bins of Figure 5 it is well calibrated and shows no strong energy trend (particularly for the standard-angle branch, where it is essentially flat), so the per-energy Ψ_{68} values are already approximate confidence intervals. We have not yet implemented per-event predictive uncertainty (e.g., the Bayesian uncertainty decomposition of Kendall and Gal [51]); such an extension would provide event-level error bars useful for source-morphology and line-search analyses and is a natural next step.

6. Conclusion

In this work, we have proposed and validated a hierarchical framework for particle trajectory reconstruction up to 90° in the BGO calorimeter with DAMPE experiment. Addressing the principal limitation of conventional methods, which fail for large-angle events due to geometric constraints, the hierarchical deep learning strategy effectively decouples the reconstruction from the auxiliary detectors (STK).

Our approach uses a "divide-and-conquer" strategy, with a high-precision SVM classifier that routes events to specialized reconstruction branches. For standard incidence angles (0°–60°), the PGHNN synergizes analytical physical priors with data-driven features, achieving an angular resolution of 1.40° with BGO data alone and superior stability against energy fluctuations compared to standard algorithms. For large incidence angles (60°–90°), where analytical methods collapse, the DLD model successfully recovers the trajectory information from truncated shower topologies, maintaining a resolution of 1.31°. Collectively, this framework achieves nearly 100% reconstruction efficiency across the entire angular domain for electrons in the 100 GeV to 1 TeV range, raising the effective acceptance of the detector by a factor of ≈ 1.8. Applied to flight electron candidates with STK tracks (30°–71°), the predicted directions agree with the STK tracks at the degree level (median 1.44°, 68th percentile 1.98°). This is a consistency check between two reconstructions: the STK track has its own finite resolution and is not a truth direction, so the comparison does not yield an absolute angular resolution of the BGO-only model.

Looking forward, this work points to several concrete research directions. First, we aim to extend the angular coverage of the reconstruction framework beyond the current 0°–90° range to include grazing and upward-going events in the 90°–130° regime. In this range, particles enter the detector from below the horizon, traversing the BGO calorimeter in the reverse direction and producing fundamentally different shower topologies characterized by asymmetric energy deposition. Extending the DLD architecture to handle these inverted geometries will further expand the effective field of view of the detector and improve the sensitivity to upward-going cosmic-ray and Earth-albedo events. Second, we aim to extend this methodology beyond the electron to include nucleon cosmic rays, such as protons, helium, and heavier nuclei (C, Mg, Fe). Reconstructing these species solely with the calorimeter is challenging due to the complexity of hadronic showers, but

it is essential for extending the spectral measurements to the PeV range where statistics are scarce. Third, we plan to explore a unified, multi-task learning architecture capable of simultaneous trajectory reconstruction and energy estimation. By sharing feature extraction backbones, such a model could not only improve computational efficiency for on-board processing but also use the mutual correlations between spatial and calorimetric features to further improve overall precision.

Data and Code availability

The authors do not have permission to share data. But the code to train and evaluate models is available at `https://github.com/kelly-struck/Electron-Trajectory-Reconstruction.git`.

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